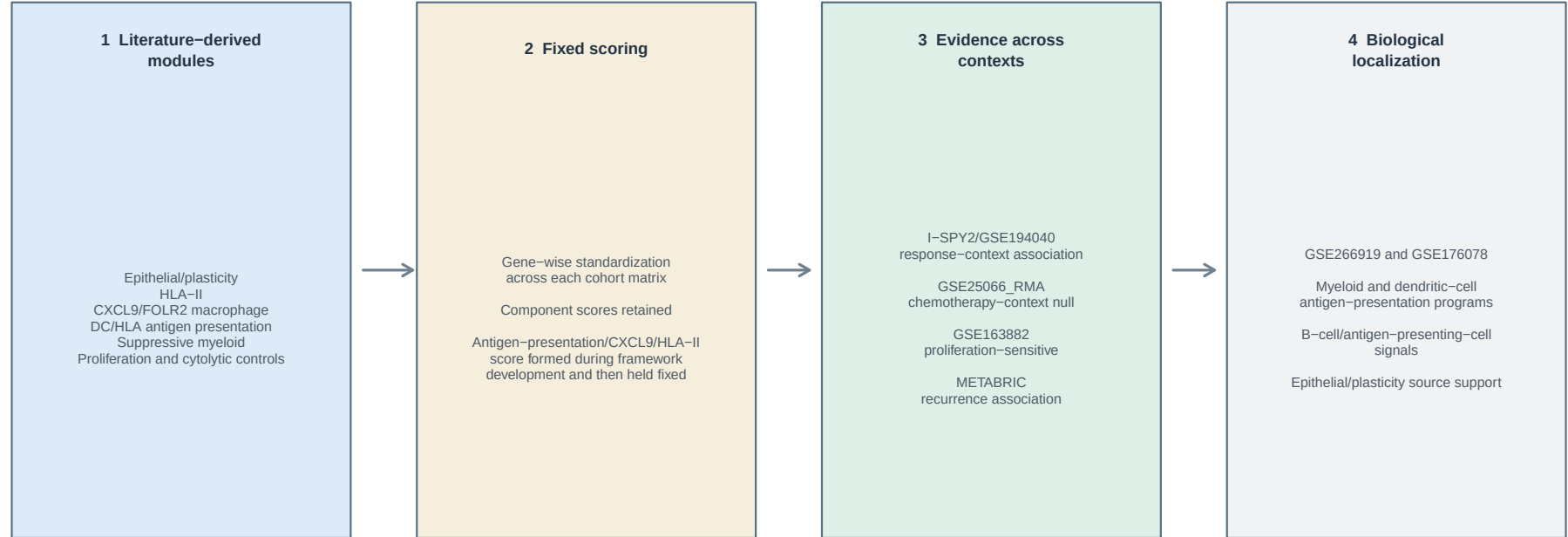


TNBC state-decomposition study design

Literature-derived gene sets; antigen-presentation/CXCL9/HLA-II score held fixed for later analyses



Interpretation boundary: transcriptomic state mapping across treatment contexts; not a general pCR assay or clinical-grade predictor.